

# Lecture 1: Computer Graphics

## What is Computer Graphics?

Computer Graphics is a field of computer science that deals with creating, manipulating, and displaying visual images using computers. These images can be 2D or 3D and are used in design, art, science, engineering, and entertainment.

## Why Do We Need Computer Graphics?

- Visualization of complex data
- User interface design (GUIs)
- Simulation and training (e.g., medical, military)
- Entertainment (movies, games, virtual worlds)
- Design and engineering (CAD/CAM)
- Educational tools and interactive content

## Important Tools in Computer Graphics

- Adobe Photoshop (2D Image Editing)
  - Use Case: Photo editing, UI design
  - Features: Layers, filters
  - Cost: Paid (subscription)
  - Advantages: Industry standard, rich features
  - Disadvantages: Expensive, learning curve
- Blender (3D Design & Animation)
  - Use Case: Movies, simulations
  - Features: Modeling, rigging, rendering
  - Cost: Free
  - Advantages: Powerful, open-source
  - Disadvantages: Complex UI
- Autodesk Maya (3D Animation)
  - Use Case: High-end effects
  - Features: Rigging, dynamics
  - Cost: Expensive
  - Advantages: Film-grade output
  - Disadvantages: Resource intensive
- Unity (Game Development)
  - Use Case: Games, VR/AR apps
  - Features: Real-time rendering, scripting
  - Cost: Free/Paid
  - Advantages: Cross-platform
  - Disadvantages: Requires coding skills

- CorelDRAW (Vector Design)
  - Use Case: Logos, print design
  - Features: Vector editing
  - Cost: Paid
  - Advantages: Good for print
  - Disadvantages: Less web integration
- AutoCAD (CAD Engineering)
  - Use Case: Architecture, blueprints
  - Features: Precision drawing
  - Cost: Paid
  - Advantages: High accuracy
  - Disadvantages: Not for art
- Figma (UI/UX Design)
  - Use Case: Web and app interfaces
  - Features: Collaborative, prototyping
  - Cost: Free/Paid
  - Advantages: Browser-based, teamwork-friendly
  - Disadvantages: Limited advanced features

### Advantages and Disadvantages of using tools

Advantages:

- High-quality visuals
- Speeds up design process
- Realistic simulations
- Content reusability
- Interactive and engaging

Disadvantages:

- Requires high-end hardware
- Steep learning curves
- Expensive software
- Time-consuming rendering

## Lecture 2: Multimedia Systems

### What is Multimedia?

Multimedia is the combination of different content forms (text, images, audio, video, and animation) integrated into one system to deliver information or entertainment interactively.

### Forms of Multimedia Data

- ✧ Text: Basic written content

- ✧ Graphics: Still images, illustrations
- ✧ Audio: Sound, music, narration
- ✧ Video: Moving images or recordings
- ✧ Animation: Simulated motion
- ✧ Hypermedia: Linked multimedia content

## Input Devices in Multimedia Systems

- ✧ **Keyboard**: A primary input device used to enter text and commands into a computer.  
Example: Used to type descriptions or instructions in a multimedia presentation.
- ✧ **Mouse**: A pointing device for selecting items and navigating interfaces.  
Example: Used to select tools or navigate in design software like Photoshop.
- ✧ **Microphone**: Captures audio input such as voice or sound.  
Example: Used to record narration or voiceovers for multimedia videos.
- ✧ **Scanner**: Digitizes physical images or documents.  
Example: Used to import hand-drawn art into a digital multimedia project.
- ✧ **Digital Camera**: Captures digital photos and videos.  
Example: Used to shoot videos or take photos for use in a multimedia application.
- ✧ **Graphics Tablet**: Used for drawing with a stylus on a sensitive surface.  
Example: Used by artists to draw directly into design software.
- ✧ **Webcam**: Captures real-time video input.  
Example: Used for recording video content or live streaming.

## Output Devices in Multimedia Systems

- ✧ **Monitor**:  
A screen that displays text, images, animations, and video.  
Example: Used to view edited images or multimedia presentations.
- ✧ **Speakers**:  
Devices that output audio signals.

Example: Used to listen to narration, music, or sound effects.

- ✧ **Printer:**  
Produces hard copies of digital documents or images.  
Example: Used to print posters or graphic designs.
- ✧ **Projector:**  
Projects images or videos onto a large screen.  
Example: Used for classroom presentations or seminars.
- ✧ **VR Headset:**  
Wearable device that delivers immersive 3D content.  
Example: Used in VR games or training simulations.

### Advanced use of Multimedia

- ✧ **Multimedia Compression:** Reduces the size of media files while preserving quality.  
Example: Used to save disk space and enable smooth streaming (e.g., MP3, JPEG, H.264).
- ✧ **Streaming Media:** Delivers audio/video content over the internet in real-time.  
Example: Watching a movie on Netflix or YouTube without downloading.
- ✧ **Virtual Reality (VR):** Creates simulated environments that users can interact with.  
Example: Used in training simulators or immersive gaming.
- ✧ **Augmented Reality (AR):** Overlays digital information on the real world.  
Example: Used in apps like Pokémon GO or interior design previews.
- ✧ **Multimedia Authoring Tools:** Software used to create multimedia projects.  
Example: Using Adobe Animate to build interactive e-learning content.
- ✧ **Interactive Multimedia:** Allows users to control content flow and respond to prompts.  
Example: Educational tutorials where students click to explore lessons.
- ✧ **Multimedia Networking:** Handles delivery of multimedia over networks.  
Example: Video conferencing with real-time video, audio, and file sharing.